🚕 Uber Fare Price Prediction (Machine Learning Project)

📌 Overview

This project focuses on predicting Uber ride fares using Machine Learning. The main objective is to analyze ride data, perform feature engineering, and build a regression model that can accurately estimate fare based on trip characteristics.

🎯 Objective

To develop a predictive model that estimates Uber fare price using features such as distance, time, and ride conditions.

📊 Dataset Description

The dataset contains information about Uber rides including pickup time, location coordinates, and fare amount.

🔑 Key Features:

- `distance in km` – Distance between pickup and drop-off locations using Phaversine formula
- `pickup_datetime` – Date and time of ride
- `passenger_count` – Number of passengers

🧠 Engineered Features:

- `hour` – Hour of the day
- `day_of_week` – Day index (0–6)
- `is_weekend` – Weekend indicator (0 or 1)
- `part_of_day` – Morning / Afternoon / Evening / Night

🧹 Data Preprocessing

- Removed invalid and zero-distance trips
- Filtered unrealistic fare values and extreme outliers
- Converted datetime into useful time-based features
- Applied one-hot encoding for categorical variables
- Ensured clean and structured dataset for modeling

📈 Exploratory Data Analysis (EDA)

- Visualized **fare vs distance** using scatter plots
- Identified strong positive correlation between distance and fare
- Analyzed impact of time-based features on pricing
- Detected skewness and outliers in the dataset

🤖 Model Used

- **Linear Regression (Scikit-learn)**

📊 Model Performance

- **R^2 Score:** 0.73
- **Mean Absolute Error (MAE):** ~2.95
- **Root Mean Squared Error (RMSE):** ~4.85

👉 The model provides a good fit and produces realistic predictions.

🔮 Sample Predictions

Distance (km)	Conditions	Predicted Fare
6 km	Evening	₹18.58
10 km	Weekend Evening	₹26.94
15 km	Night	₹37.23

🧠 Key Insights

- Distance is the **most important feature** affecting fare
- Fare increases approximately linearly with distance
- Time-based features have moderate influence
- Passenger count and weekend have minimal impact
- Data cleaning significantly improves model accuracy

🚀 Future Improvements

- Implement advanced models (Random Forest, XGBoost)
- Include real-world factors like traffic and surge pricing
- Deploy the model using Streamlit or Flask
- Improve feature engineering for better accuracy

🛠️ Tech Stack

- Python
- Pandas, NumPy
- Matplotlib, Seaborn
- Scikit-learn

📌 Conclusion

This project demonstrates a complete machine learning workflow, including data preprocessing, feature engineering, visualization, model building, and evaluation. The Linear Regression model achieved good performance and successfully predicts Uber fares based on input features.

📂 Project Structure

```
Uber-Fare-Prediction/
│
├── Uber_Fare_Price_Prediction.ipynb
├── README.md
```

⭐ Author

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📎 Note

This project is built for learning and demonstration purposes as part of a Machine Learning journey.